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Investigating the molecular mechanisms between type 1 diabetes and mild cognitive impairment using bioinformatics analysis, with a focus on immune response

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ABSTRACT

The immune system is involved in the development and progression of several diseases. Type 1 diabetes mellitus (T1DM), an autoimmune disorder, influences the progression of several other conditions; however, the link between T1DM and mild cognitive impairment (MCI) remains unclear. This study investigated the underlying immune response mechanisms that contribute to the development and progression of T1DM and MCI. Microarray datasets for MCI (GSE63060) and T1DM (GSE30208) were retrieved from the Gene Expression Omnibus database. Differentially expressed genes (DEGs) were identified using the limma package. To explore the functional implications of these DEGs, Gene Ontology and Kyoto Encyclopedia of Genes and Genomes pathway enrichment analyses were conducted using ClusterProfiler. Protein–protein interaction networks for the DEGs were constructed using the STRING database and visualized using Cytoscape. The Molecular Complex Detection algorithm was used to analyze DEGs. Immune cell infiltration in patients with T1DM and MCI was analyzed using the xCell method. Gene set enrichment analysis was used to gain in-depth insights into the functional characteristics of T1DM and MCI. Immune-related genes were obtained from the GeneCards and ImmPort databases. Machine learning algorithms were used to identify potential hub genes associated with immunity for T1DM and MCI diagnosis, and the diagnostic value was assessed using the receiver operating characteristic curve. The identified genes were validated using quantitative polymerase chain reaction. In the T1DM and MCI datasets, 610 DEGs showed consistent trends, of which 232 and 378 were upregulated and downregulated, respectively. Immune response analysis revealed significant changes in the immune cells associated with MCI and T1DM. Using immune-related genes, DEGs, and machine learning techniques, we identified *CD3D* in the MCI and T1DM groups. We observed a gradual decline in the cognitive function of mice with T1DM as the disease progressed. *CD3D* expression increased with increasing disease severity; *CD3D* primarily affected CD4+ T cells. This study revealed a complex interaction between T1DM and MCI, providing novel insights into the intricate relationship between immune dysregulation and cognitive impairment and expanding our understanding of these two interconnected disorders. These findings will facilitate the development of therapeutic interventions and identification of potential therapeutic targets.

Abbreviations: AMPK, AMP-activated protein kinase; AUC, area under the curve; DEGs, differentially expressed genes; DM, diabetes mellitus; DT, decision tree; GBM, Gradient Boosting Machine; GO, Gene Ontology; GSEA, gene set enrichment analysis; GSVA, Gene Set Variation Analysis; KEGG, Kyoto Encyclopedia of Genes and Genomes; LASSO, least absolute shrinkage and selection operator; MCI, mild cognitive impairment; MCODE, Molecular Complex Detection; NC, normal control; PCA, principal component analysis; PCR, polymerase chain reaction; PPI, protein–protein interaction; RA, rheumatoid arthritis; RF, random forest; ROC, receiver operating characteristics; STZ, streptozotocin; SVM, support vector machine; T1DM, type 1 diabetes mellitus.

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1. Introduction

Diabetes mellitus (DM) is characterized by hyperglycemia owing to impaired insulin secretion, insulin action, or both. Type 1 DM (T1DM) occurs more commonly in adolescents and is associated with B cell destruction in the pancreatic islets, resulting in insulin production deficiency and elevated blood glucose levels [1], with subsequent severe effects on multiple organs. Recently, mild cognitive impairment (MCI) has been identified as a significant DM comorbidity [2]. Animal models and human studies have linked T1DM to diminished performance in various cognitive functions, culminating in dementia [3]. A 20-year longitudinal study reported a 19 % exacerbation in cognitive impairment in diabetes patients during midlife than in individuals without diabetes [4]. A retrospective study reported that the dementia risk ratio for hospitalized T1DM patients was 1.65 times higher than that of individuals without diabetes [5]. These findings necessitate the exploration of the molecular mechanisms underlying the co-pathogenesis of T1DM and MCI. T1DM and its associated complications markedly affect cognitive functions, often leading to cognitive impairment [6,7]. Hence, it is imperative to emphasize the importance of prevention and early intervention against cognitive decline.

Cognitive deficits can be categorized into three stages of increasing severity: diabetes-related cognitive impairment, MCI, and dementia. Notably, poorly controlled blood sugar levels may elevate the risk of developing dementia, a connection so significant that some researchers have dubbed Alzheimer's as "diabetes of the brain" or "type 3 diabetes" [8]. Given that MCI often precedes the onset of dementia, understanding its underlying pathophysiology is crucial. From a pathophysiological perspective, T1DM substantially heightens the risk of MCI owing to hyperglycemia, insulin resistance [9,10], and disrupted energy metabolism [11], among other factors. Presently, studies investigating T1DM-related MCI have predominantly focused on diabetic ketoacidosis and microangiopathy [12]. Consequently, immune factors have received less attention in the context of T1DM-related MCI, despite the association between T1DM and MCI with inflammation and immune responses. Therefore, exploring the role of immune response in T1DM and MCI is imperative to identify early diagnostic markers and potential therapeutic targets for dementia.

This study investigated the underlying immune response mechanisms contributing to the development and progression of T1DM and MCI.

Previous studies have extensively investigated the molecular biomarkers of Alzheimer's disease(AD), relying primarily on magnetic resonance imaging [13,14] and cerebrospinal fluid analysis [15], which present challenges stemming from clinicopathological heterogeneity and invasiveness. In this study, we conducted an integrated bioinformatics analysis to elucidate the potential molecular mechanisms underlying the development of T1DM complicated by MCI. Our multiomic data integration provides insights into pathological mechanisms linking dysregulated immunity to cognitive decline in T1DM. These findings will aid in the development of strategies to prevent or delay cognitive impairment in this patient population.

2. Materials and methods

2.1. Data collection and processing

The gene expression profiles of peripheral whole blood were obtained from the National Centre for Biotechnology Information Gene Expression Omnibus public database (GEO, https://www.ncbi.nlm.nih.gov/geo/). We used the GSE63060 dataset from a large cross-European AD biomarker study and the EU-funded AddNeuroMed Cohort that contained RNA samples obtained from peripheral blood. This dataset included the data of 98 healthy controls and 78 patients with MCI. To ensure data quality, we used the principal component analysis (PCA) algorithm to process the data and identified several outliers. Outliers GSM1539171, GSM1539113, GSM1539389, GSM1539155, GSM1539184, GSM1539331, GSM1539180, and GSM1539149 were subsequently removed. Additionally, we utilized the GSE30208 gene expression profile, which contains 63 peripheral blood RNA samples from individuals with T1DM.

2.2. PCA

To eliminate sample outlier biases, we performed PCA of gene expression for each sample in mild cognitive impairment and type 1 diabetes group. Gene expression was normalized for each sample (Supplementary Fig. 1A,B), and PCA was completed on these data using the factoextra package; for all subsequent analyses, samples outside the 95 % confidence ellipse in their respective disease were excluded (Supplementary Fig. 1C, D).

2.3. Identification of differentially expressed genes (DEGs)

In the MCI and T1DM groups, DEGs were identified using the "limma" package in R [16]. The statistical significance threshold for differential gene expression was set at adj-p-value < 0.05 and $|\log2 \text{ fold change}| > 0$ [17,18]. A volcano plot, generated with the "ggplot2″ package in R, was used to visually represent the total DEGs in the MCI and T1DM groups. Using the "Pheatmap" package in R, a volcano plot was generated to identify the top 10 most significant upregulation and downregulation DEGs (totaling 20 DEGs). Venn diagrams were used to visualize and identify the common immune-related DEGs among immune genes from the publicly available GeneCards (https://www.genecards.org/) and ImmPort (https://www.immport.org/home) databases.

2.4. Protein–protein interaction (PPI) network construction and hub gene analysis

To investigate the interactions between DEGs, PPI network analysis was conducted using the online tool STRING [19] (version 11.5), with a combination threshold set at >0.4 [20]. The interaction information was imported into Cytoscape (version 3.9.1) (https://cytoscape.org/) [21], and the degree algorithm of the Cytoscape software was used to screen hub genes. Molecular complex detection (MCODE) was applied to Cytoscape to identify functional clusters of genes in the PPI network [22].

2.5. Gene set enrichment analysis (GSEA)

Gene Ontology (GO) categorical analysis is a commonly employed method for conducting bioinformatic analyses on large datasets. The Kyoto Encyclopedia of Genes and Genomes (KEGG) is a valuable database for understanding the overall functions and utility of biological systems. For the analysis, the R package ClusterProfiler [23,24] was used to facilitate GO and KEGG analyses and the cluster profile and Gene Set Variation Analysis (GSVA) packages were used to explore specific signaling pathways. Gene sets used in the analysis were obtained from MSigDB (c5.go.bp.v7.4. symbols.gmt) [25]. GSEA was used to detect potential pathways associated with the gene expression matrix.

2.6. Immune cell infiltration assessment

The gene expression datasets GSE63060 and GSE30208 were used for immune cell infiltration analysis using the R packages CIBERSORT and xCell [26]. CIBERSORT, a deconvolution algorithm, was used to integrate the annotated genomes of diverse immune cell subpopulations to calculate the relative abundance of 22 immune cell types within tissues.

Non-parametric correlations (Spearman's rank correlation) were used to ascertain the relationship between the three core biomarkers and immune-infiltrated cells [27].

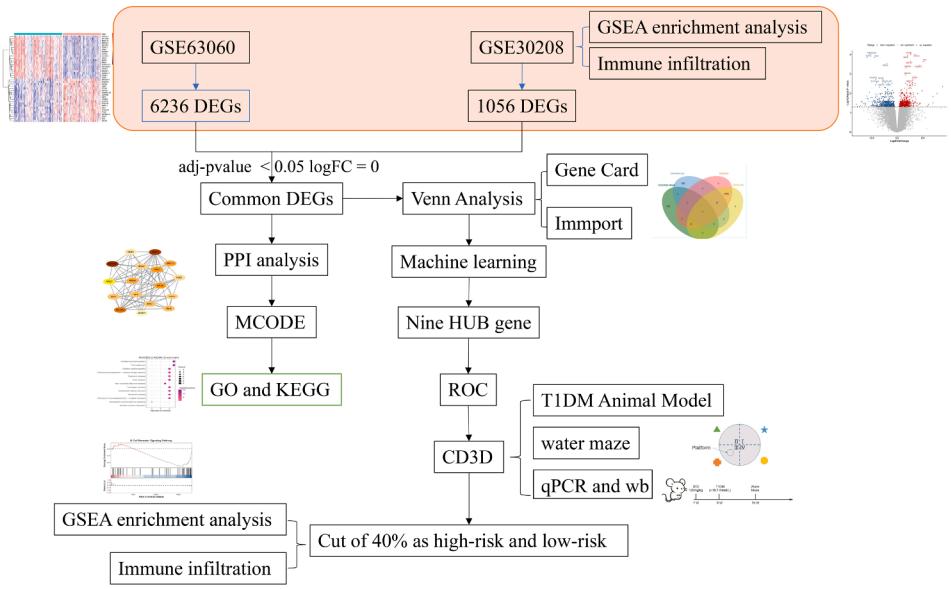


Chart 1. The flow chart of this study.

2.7. Identification of immune-related hub genes

Nine commonly used machine learning algorithms, namely the Elastic Net(Enet) Regression, Support Vector Machine (SVM) [28], Random Forest (RF) [29], Recursive Partitioning(rpart) Regression Trees, Gradient Boosting Machine (GBM) [30], Ridge Regression, least absolute shrinkage and selection operator (LASSO) [31], Generalized Linear Model (GLM), and Partial Least Squares Regression Generalized Linear Model (plsRglmmodel), were used to screen disease candidate genes. Enet, Ridge, LASSO, and GLM were implemented using the "glmnet" package in R software, whereas rpart was performed using the "rpart" package in R software. plsRglmmodel was performed with the "plsRglm" package, the RF algorithm utilized the "randomForest" package, and GBM was implemented using the "gbm" package in R software. The SVM-RFE algorithm utilized the "e1071," "kernlab," and "caret" packages in R software. Finally, an Upset plot was used to visually represent the candidate disease genes obtained from the results of the nine machine learning algorithms.

2.8. Establishment of a receiver operating characteristic

The receiver operating characteristic (ROC) curve is a tool used to visualise the discriminatory power of a predictive model by plotting a curve illustrating the correlation between true positive and false positive rates, which helps to assess the classification validity of the model at different thresholds. The diagnostic value of the hub genes was evaluated with ROCs, internal validation of the model was performed using bootstrap resampling, and the calibration plot is shown graphically.

2.9. Animal experiments

All animal experiments were approved by the National Ethical Committee for the Care and Use of Animals (ethical approval number: No. SCXK2020-030 was approved on January 10, 2021; animal quality certificate number: 51203500018786). Stringent measures were followed to minimize animal suffering and the number of animals used. C57BL/6 wild-type mice aged 8–10 weeks were housed in a facility at Kunming Medical University under specific pathogen-free conditions. The cages were maintained under controlled temperature and humidity conditions, with a 12-h light–dark cycle. The mice were given unrestricted access to food and water [32,33].

2.9.1. Mice model of T1DM with MCI

C57BL/6 mice were fed a normal diet for one week before intraperitoneal injection of a single dose of 150 mg/kg streptozotocin (STZ, Sigma-Aldrich) [34]. After 1 week, mice with a fasting blood glucose level of ≥16.5 mmol/l were classified as having T1DM. The control mice used in the STZ-related assays were sex- and age-matched C57BL/6 mice injected with an equivalent volume of vehicle buffer (sodium citrate buffer, pH 4.5). Subsequently, the mice in the T1DM were subjected to water maze experiments, and impaired memory was observed after 15 weeks.

2.9.2. Morris water maze

The water maze apparatus consisted of a pool measuring 110 cm in diameter, filled with clear water maintained at a temperature of 22 ± 1 °C. Within the pool, a platform with a diameter of 10 cm was submerged 1.0 cm below the water surface. During the training trials, the mice were given 60 s to swim and locate the hidden platform. Once the platform was located, the mice were allowed to remain on it for 10 s. If a mouse failed to locate the platform within the allotted time, it was guided to the platform. The mice underwent training thrice a day for five consecutive days, with 3–4-h intervals between each trial. After 24 h following the last training trial, the mice underwent a probe trial to assess memory retention. The probe trial involved removing the platform from the pool and recording the performance of the mice using a video camera (Sony, Tokyo, Japan) mounted above the maze. The video recordings were automatically processed using a video tracking system.

2.10. Quantitative real-time polymerase chain reaction (PCR) analysis

Gene expression analysis was performed for the *CD3D* expression by synthesizing cDNA using the Prime Script RT reagent kit (Takara, Kusatsu, Japan) following the manufacturer's instructions. Real-time PCR was conducted using SYBR Premix Ex Taq II (Takara, Kusatsu, Japan) according to the manufacturer's instructions. For normalization, β-actin was used as the endogenous control. The relative expression levels of mRNAs were determined using the $2^{-\Delta\Delta CT}$ (cycle threshold) method. The specific primer sequences used were *CD3D* (F: 5′-AGCGGGATTCTGGCTAGTCT-3′; R: 5′-CGCTGGTATTGCAGGTCACAA-3′) and Actb (F: 5′-GGCTGTATTCCCCTCCATCG-3′; R: 5′-CCAGTTGGTAACAATGCCATGT-3′).

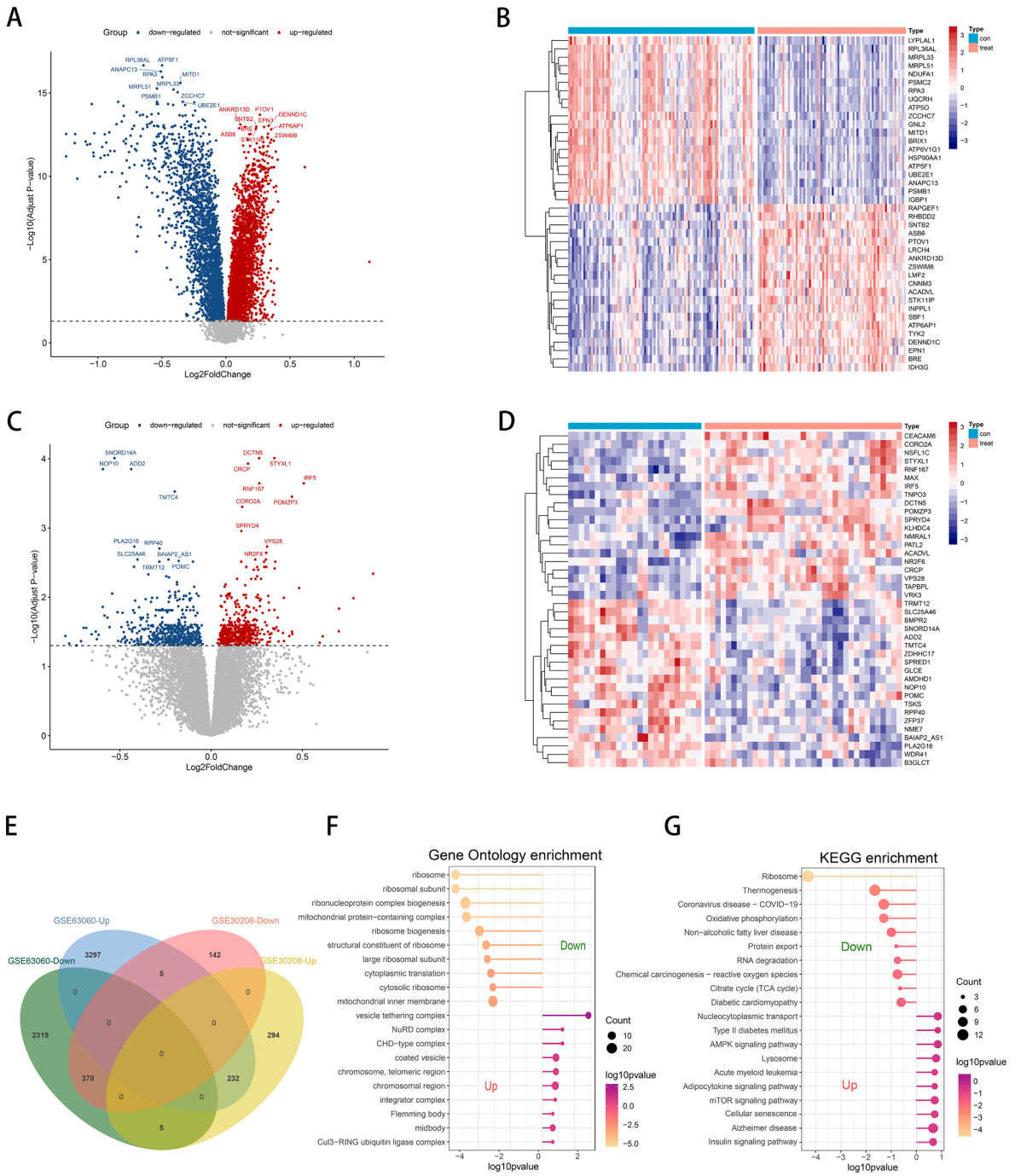


Fig. 1. Differential expression analysis of T1DM and MCI. (A, B) Volcano plot and Heatmap illustrating the differentially expressed genes in the GSE63060 dataset between the MCI and control groups. (C, D) Volcano plot and Heatmap showing the differentially expressed genes in the GSE30208 dataset between the T1DM and control groups. (E) Venn diagram depicting the overlap of differentially expressed genes in the MCI and T1DM datasets. (F, G) GO (cellular components, biological process, and molecular function) and KEGG enrichment analyses of genes commonly upregulated and downregulated in the T1DM and MCI groups. T1DM, type 1 diabetes mellitus; MCI, mild cognitive impairment.

2.11. Statistical analyses

Data were expressed as the mean ± SEM of at least three independent experimental results. Statistical analysis was performed using unpaired Student's *t*-test (two-tailed) for two groups, Student's *t*-test, or one-way analysis of variance with Bonferroni's correction for three or more groups to evaluate statistical significance using GraphPad Prism software (version 8.0). Statistical significance was set at $P < 0.05$.

2.12. Study flow chart

The detailed study flow chart (see Chart 1).

3. Results

3.1. Identification of DEGs in patients with T1DM and MCI

To identify DEGs in the controls and patients with T1DM with MCI, we used the R package "limma" (adjusted $p < 0.05$ and |log2 Fold

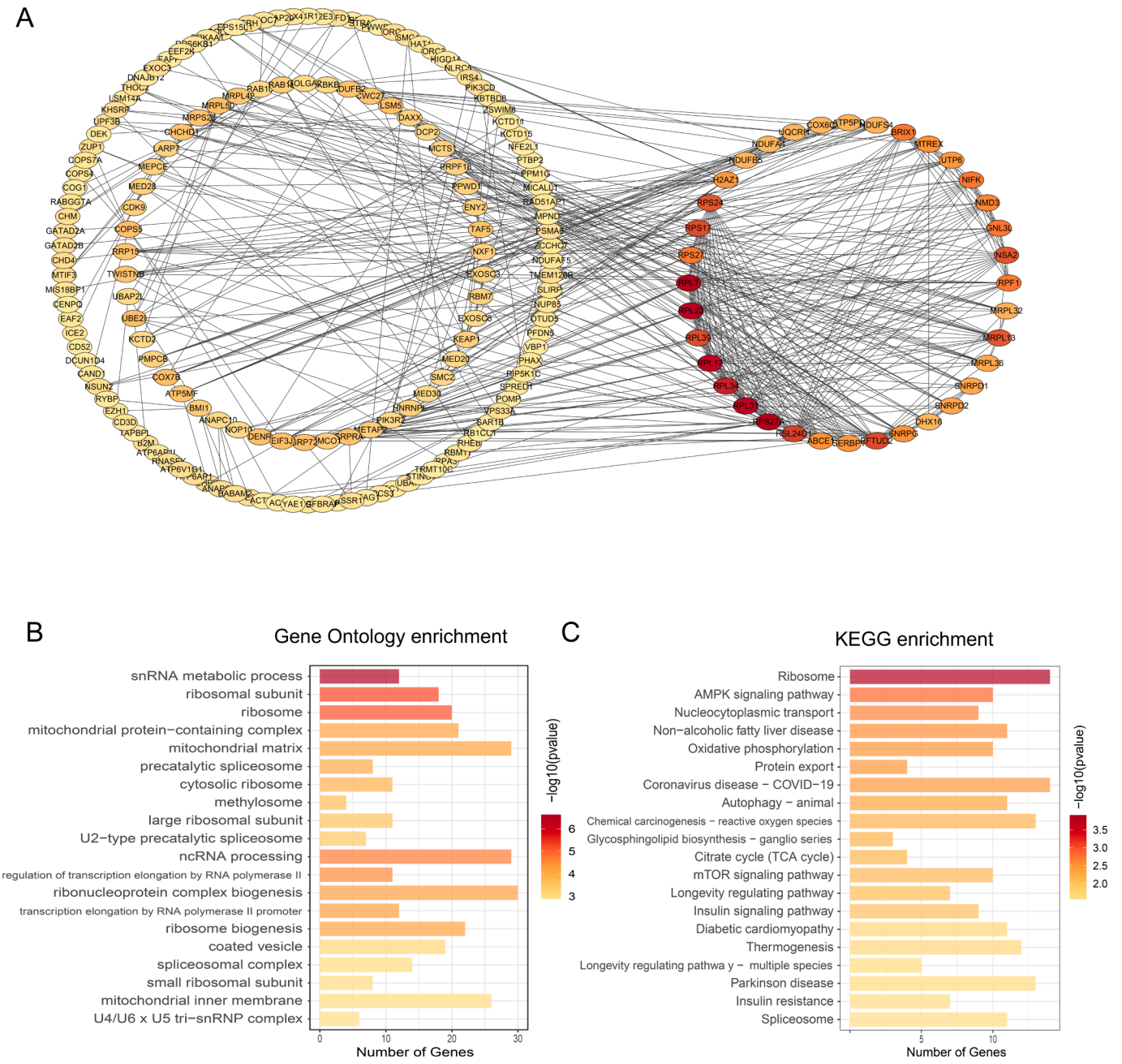


Fig. 2. Analysis of DEG PPI network and function enrichment analysis. (A) PPI network constructed with the DEGs. (B, C) GO and KEGG function enrichment analysis of DEGs. The color represents the correlation; the closer to red, the higher the significance. DEG, differentially expressed gene; PPI, protein–protein interaction.

change| > 0). In patients with MCI, compared with that seen in normal controls, we identified 6,236 DEGs, of which 3,534 and 2,702 were upregulated and downregulated, respectively (Fig. 1A). In T1DM patients and normal controls, we detected 1,056 DEGs, of which 531 and 525 were upregulated and downregulated, respectively (Fig. 1C). The top 20 DEGs with upregulated and downregulated expressions in both groups were visualized using heatmaps (Fig. 1B and D). A Venn diagram analysis of genes exhibiting a similar expression trend among the DEGs in both datasets indicated that 232 and 378 genes were commonly upregulated and downregulated, respectively (Fig. 1E). Subsequently, we conducted a functional enrichment analysis to elucidate the underlying mechanisms of DEGs with consistent expression trends. Fig. 1F displays the top 10 GO terms for each classification. Our results indicated that DEGs with upregulated expressions were notably enriched in the "vesicle tethering complex," "NuRD complex," and "CHD-type complex," whereas DEGs with downregulated expressions were primarily enriched in "ribosome," "ribosomal subunit," and "ribosome biogenesis." Fig. 1G shows the KEGG pathways enriched in these DEGs. The upregulated DEGs were primarily associated with "Nucleocytoplasmic transport" and "Type II diabetes mellitus," whereas the downregulated DEGs were predominantly enriched in "Ribosome," "Thermogenesis," and "Oxidative phosphorylation" pathways. These findings suggest that T1DM and MCI share a common pathogenesis.

3.2. Construction and analysis of the PPI network

To explore PPIs, the STRING database was used to construct a PPI network, which was visualized using Cytoscape software (Fig. 2A).

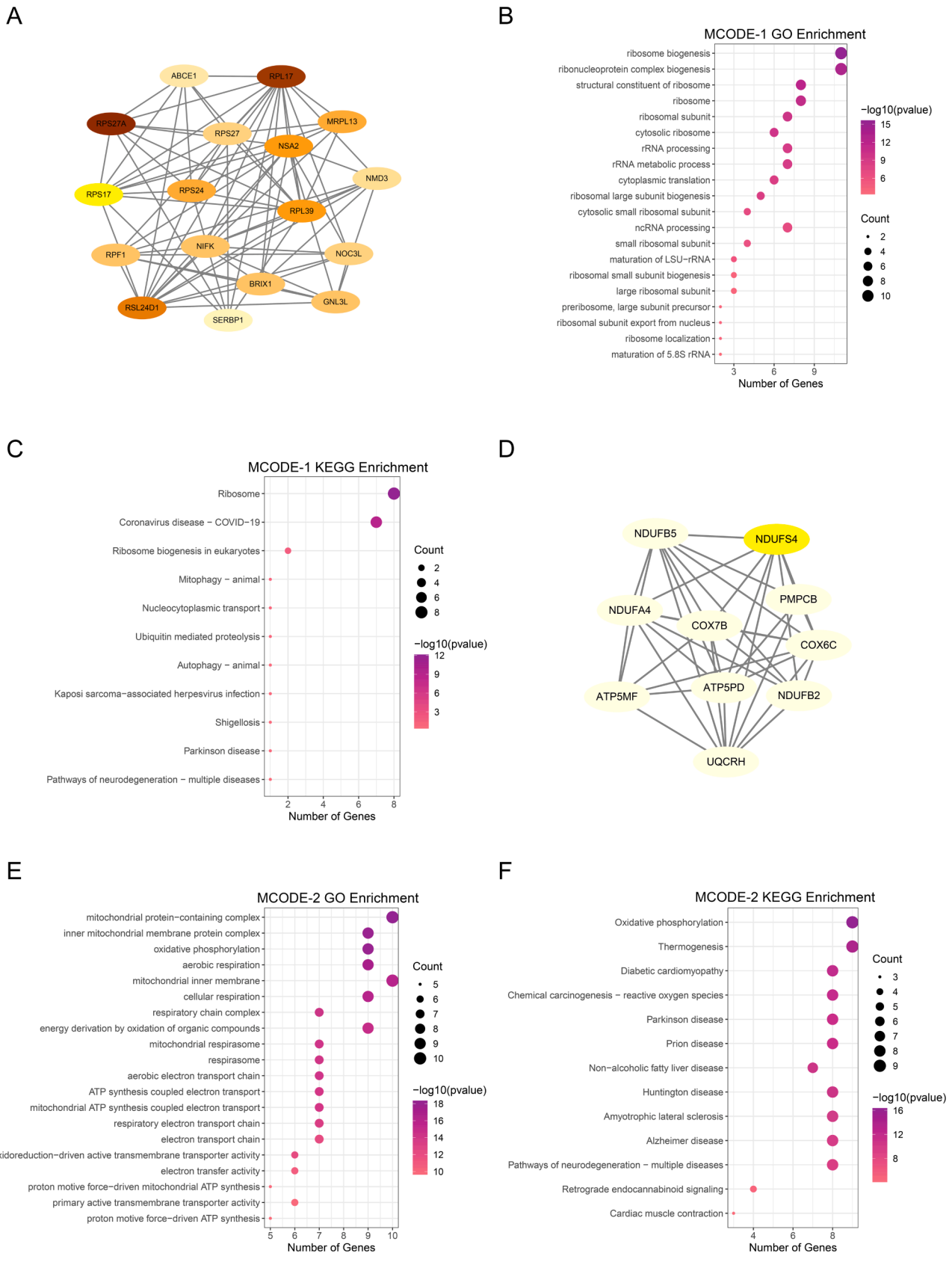


Fig. 3. MCODE analysis of the PPI module and the functional enrichment analysis of modules 1 and 2. (A, D) The TOP2 module was identified from the PPI network using the MCODE method. (B, C) The functional networks/pathways associated with the genes in the GO and KEGG enrichment analysis of module 1. (E, F) GO and KEGG enrichment analysis of module 2. The point size indicates the number of genes associated with significantly related items, and the color reflects the correlation. The closer to red, the higher the significance. PPI, protein–protein interaction.

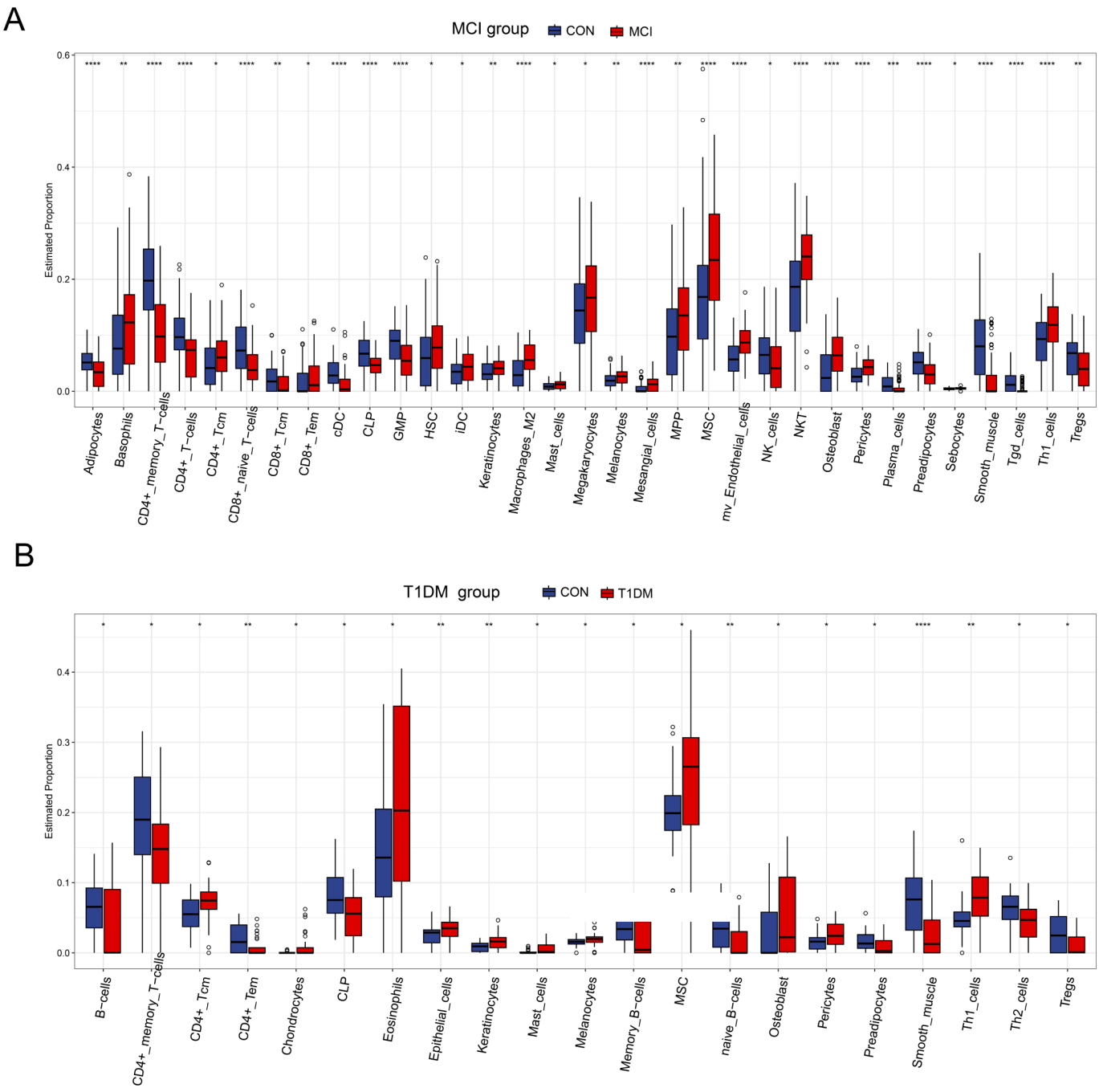


Fig. 4. Distribution of immune cell infiltration in each sample. (A) Differential analysis of immune-infiltrating cells in the MCI group using xCell analysis. (B) Differential analysis of immune-infiltrating cells in the T1DM group using xCell analysis. *p < 0.05; **p < 0.01; ***p < 0.001. T1DM, type 1 diabetes mellitus; MCI, mild cognitive impairment.

Based on GO function and KEGG pathway analyses, these genes were principally associated with RNA translation, ribosomal, diabetic cardiomyopathy, and the AMP-activated protein kinase signaling pathways (Fig. 2B,C). Subsequent analysis using the MCODE plugin within Cytoscape identified two significant modules (Fig. 3A and D). GO analysis revealed that these modules were primarily associated with ribosome biogenesis, cytoplasmic translation, adenosine triphosphate synthesis, and electron transport. Furthermore, KEGG pathway enrichment analysis indicated that these two modules were enriched in the ribosomal and oxidative phosphorylation pathways (Fig. 3B, C, E, and F).

3.3. Estimation of immune cell infiltration in MCI and T1DM

xCell analysis was used to assess variations in immune cell infiltration in the MCI and T1DM groups. The results revealed significant differences in immune cell infiltration patterns between these two conditions and healthy controls. As depicted in Fig. 4A and B (only groups with statistically significant differences are presented), CD4+ Tcm, keratinocytes, mast cells, melanocytes, MSC, osteoblasts, pericytes, and Th1 cells exhibited substantial increases in the MCI and T1DM groups. Conversely, the abundances of CD4+ memory T cells, CLP, smooth muscle cells, and Tregs significantly decreased in the MCI and T1DM groups. Remarkably, the trends in immune cell expression in both

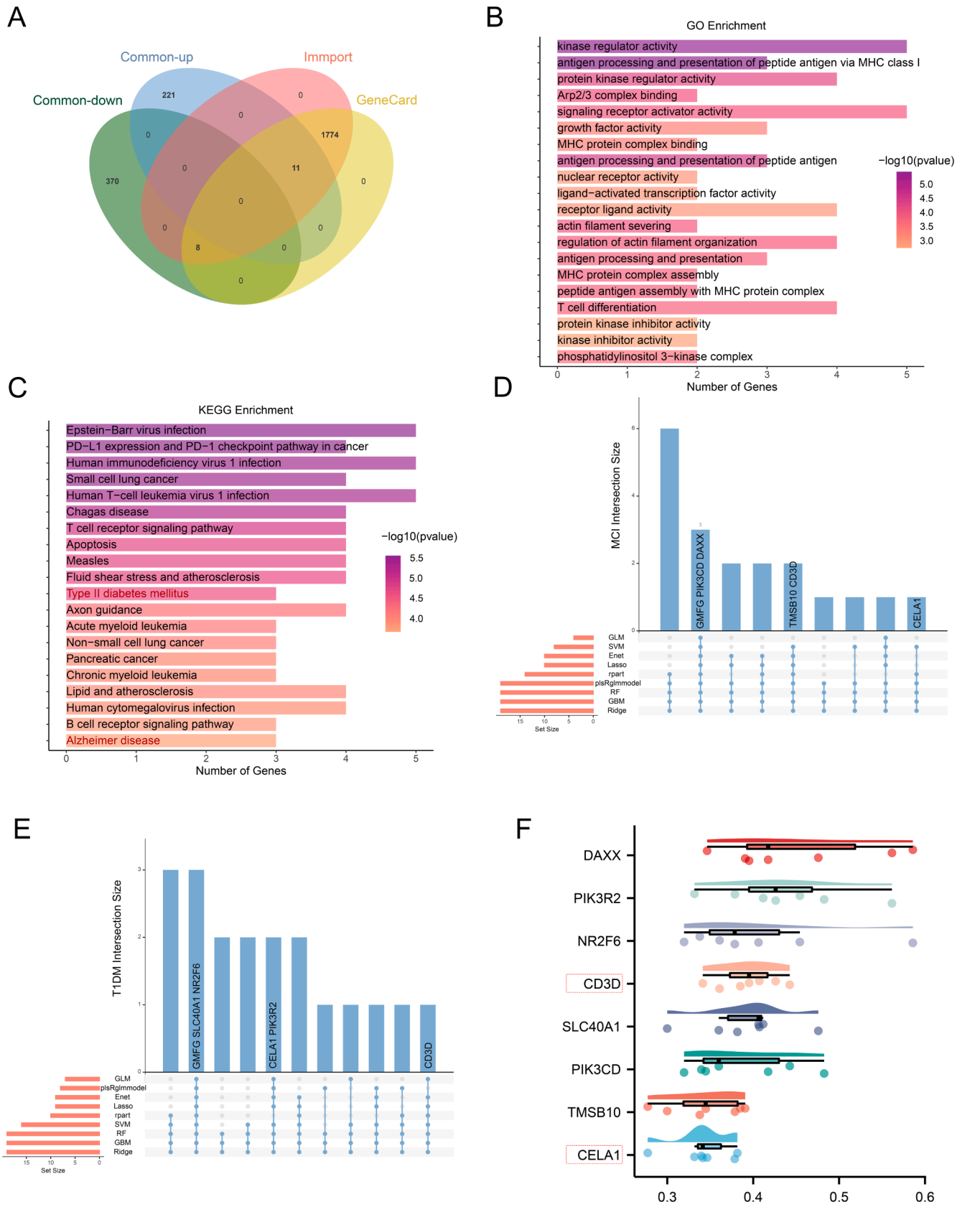


Fig. 5. Machine learning in screening candidate diagnostic biomarkers. (A) Venn diagram of the overlap of differentially expressed genes of immune-related genes. (B, C) The top 20 functional networks/pathways associated with the genes in the GO and KEGG enrichment analysis of immune-related genes. (D, E) UpSetPlot showing that the candidate diagnostic genes are identified via the above nine algorithms. (F) Friends' analysis of candidate diagnostic genes.

Table 1
Genes screened using machine learning.

	GSE63060	GSE30208
Enet	GMFG, CD3D, TMSB10, CELA1, PIK3CD, LRSAM1, DAXX, PIK3R2, NR2F6, ROBO3	GMFG, CD3D, TMSB10, CELA1, SLC40A1, DAXX, PIK3R2, NR2F6, ROBO3
GLM	GMFG, CELA1, PIK3CD, DAXX	GMFG, CD3D, CELA1, SLC40A1, PIK3R2, NR2F6, HDGF
SVM	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, PIK3CD, DAXX	GMFG, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, PIK3R2, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF
plsRglmmodel	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF	GMFG, B2M, GMFB, TMSB10, CELA1, SLC40A1, PIK3R2, NR2F6
RF	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF
rpart	GMFG, B2M, CD3D, TMSB10, BMPR2, SLC40A1, PIK3CD, DAXX, IL16, PIK3R2, RXRB, TAPBPL, ROBO3, HDGF	GMFG, CD3D, GMFB, BMPR2, SLC40A1, LRSAM1, DAXX, NR2F6, TAPBPL, ROBO3
GBM	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF
Ridge	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF	GMFG, B2M, CD3D, GMFB, TMSB10, CELA1, BMPR2, SLC40A1, PIK3CD, LRSAM1, DAXX, IL16, PIK3R2, RXRB, IKBKB, NR2F6, TAPBPL, ROBO3, HDGF
LASSO	GMFG, CD3D, TMSB10, CELA1, PIK3CD, LRSAM1, DAXX, PIK3R2, NR2F6, ROBO3	GMFG, CD3D, TMSB10, CELA1, SLC40A1, DAXX, PIK3R2, NR2F6, ROBO3

diseases were highly consistent. Furthermore, we conducted a correlation analysis of immune cells in these two diseases using CIBERSORT (Supplementary Fig. 2A and B). These findings underscore the significant similarity in the immune cell infiltration profiles of these two diseases, suggesting that the immune response plays an important role in the development of T1DM-related MCI.

3.4. Machine learning-based selection of immune-related genes in T1DM and MCI

Immune-related genes were retrieved from the Import and GeneCards databases, and a Venn diagram-based analysis was conducted using the DEGs, which showed consistent expression trends in T1DM and MCI. As illustrated in Fig. 5A, 19 genes—8 downregulated and 11 upregulated—were identified. Furthermore, we examined the correlation between the immune-related genes and immune cells (Supplementary Fig. 2C and D). The 19 genes were subjected to GO and KEGG enrichment analyses, which indicated that kinase regulator activity, antigen processing, presentation of peptide antigens via major histocompatibility protein class I, protein kinase regulator activity, and Arp2/3 complex binding were the most highly enriched categories in the GO analysis of these central genes (Fig. 5B). KEGG analysis (Fig. 5C) revealed the involvement of these genes in crucial pathways, such as the T cell receptor signaling pathway, T2DM, and AD. To identify potential diagnostic biomarkers, we used the following nine distinct algorithms: Enet, SVM, RF, rpart, GBM, Ridge, LASSO, GLM, and plsRglmmodel. The seed parameter was set to 11 for MCI and T1DM groups. In the MCI group, the feature variables *GMFG, PIK3CD, DAXX, TMSB10, CD3D,* and *CELA1* were identified (Fig. 5D), whereas in the T1DM group, *GMFG, SLC40A1, NR2F6, CELA1, PIK3R2,* and *CD3D* were identified (Fig. 5E). Table 1 lists all genes screened using machine learning. *CD3D, CELA1,* and *GMFG* were selected for MCI and T1DM diseases. The Friends analysis was conducted on the nine genes identified using algorithms for T1DM and MCI (Fig. 5F); the results showed that *CD3D* had the highest score. Unfortunately, *GMFG* was not detected in the Friends analysis, which indicated that it was not co-enriched with other genes in the GO analysis. These results suggest the potentially pivotal role of *CD3D* in the development of T1DM-related MCI.

3.5. Diagnostic value assessment

A ROC curve was constructed to assess the significance of the top nine genes: *DAXX, PIK3R2, NR2F6, SLC40A1, PIK3CD, TMSB10, CD3D, CELA1*, and *GMFG* (Fig. 6). For each of these genes, we calculated the area under the curve (AUC) and the corresponding 95 % confidence intervals. To evaluate the diagnostic specificity and sensitivity of each gene, 30 % of the T1DM (GSE32028) and MCI (GSE63060) datasets were designated as the training set, with the remaining 70 % as the validation set. In the training set, *CD3D* showed a mean AUC of 0.823 for T1DM and MCI, which surpassed those of the other genes. Similarly, in the test set, *CD3D* showed a mean AUC of 0.738 for T1DM and MCI, which was significantly higher than those for the other genes.

3.6. Validation of CD3D gene expression

Next, we established a T1DM mouse model and used the water maze test to assess the learning and memory function of mice aged 15 weeks. Fig. 7A shows the experimental setup. As shown in Fig. 7B, mice with T1DM had significantly lower learning and memory abilities than the normal control (NC) from day 3, and the latent period (in seconds) was lower in the NC group.

In Fig. 7C–E, on day 6, compared with the T1DM group, the NC group displayed significantly reduced latency, and the average time and percentage of distance covered by the mice in the NC group in the quadrant where the platform was located were significantly higher than those of T1DM mice. After hiding the platform, the mice in the NC group spent significantly more time searching for the platform with increased frequency than mice with T1DM (Fig. 7F), suggesting that mice with T1DM possess markedly impaired learning and memory compared with their NC counterparts. Finally, we utilized qPCR (Fig. 7G) and western blotting (Fig. 7H,I) to detect the expression of *CD3D* in the blood of mice with T1DM; the results demonstrated a significant increase in both the mRNA and protein levels of *CD3D*.

3.7. Single-gene GSEA of diagnostic genes

To explore the underlying biological mechanisms for the differences between the high- and low-risk groups, we conducted a single-gene GSEA to investigate the role of *CD3D* in the T1DM (Fig. 8A) and MCI groups (Fig. 8B). Based on the expression of *CD3D*, we divided the top 40 % into the high-risk group and the bottom 40 % into the low-risk group. This showed that in both disease groups, *CD3D* significantly participated in key biological processes, including ribosome biosynthesis, oxidative phosphorylation, the B cell receptor signaling pathway, AD, Parkinson's disease, and Huntington's disease. This finding underscores the pivotal role of *CD3D* in energy metabolism and neurodegenerative diseases.

3.8. Immune cell infiltration analysis of CD3D gene

Next, we employed xCell analysis to evaluate immune cell infiltration variations between the MCI and T1DM groups. As illustrated in

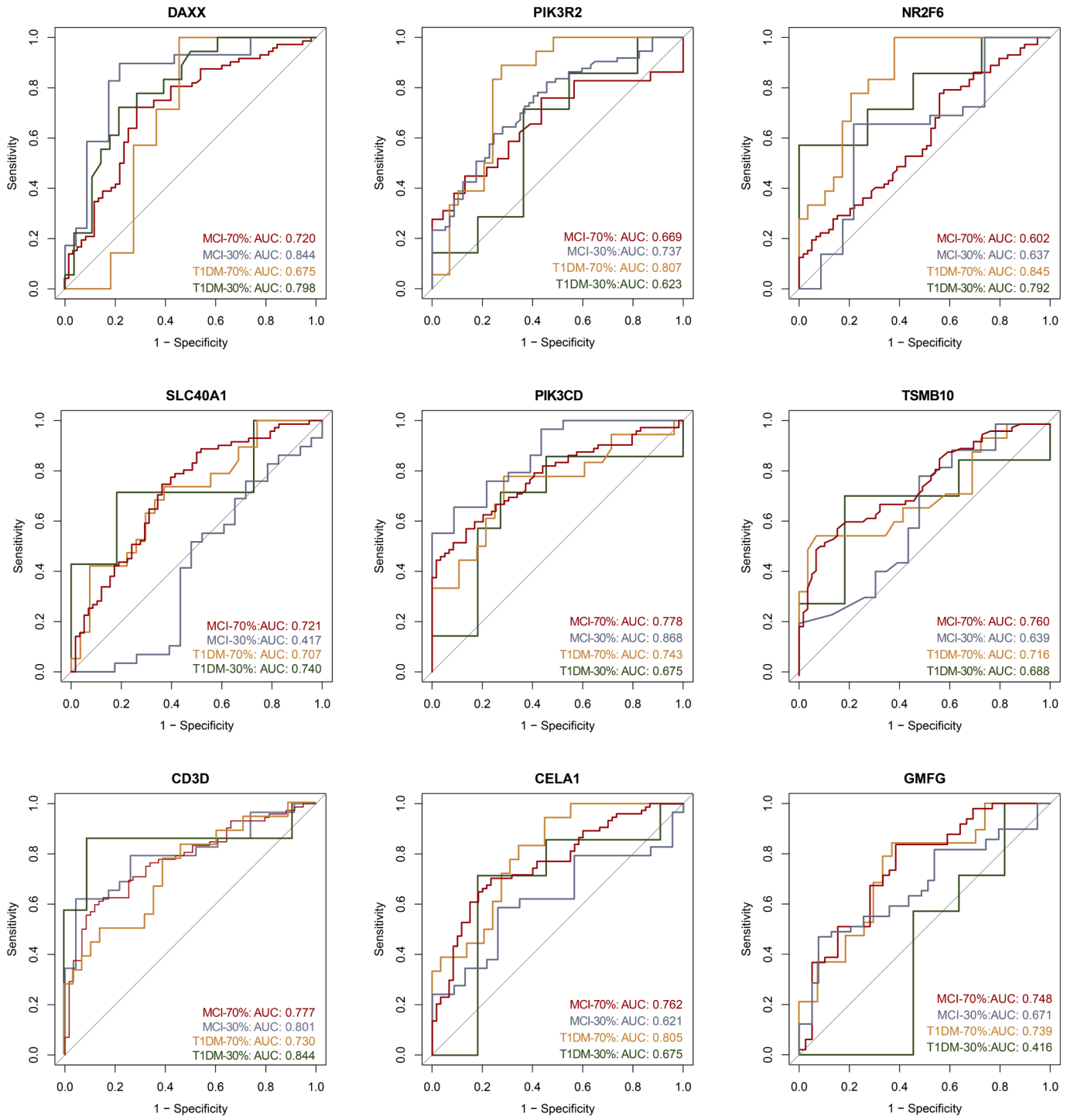


Fig. 6. ROC curve of candidate diagnostic biomarkers. The ROC curve of each candidate gene (DAXX, PIK3R2, NR2F6, SLC40A1, PIK3CD, TMSB10, CD3D, CELA1, and GMFG). AUC, area under the curve.

Fig. 9A, we focused on immune cells significantly associated with *CD3D* (correlation coefficient > 0.3 and $p < 0.05$) and changes in T1DM. *CD3D* expression positively correlated with cDC, CD4+ T cells, CD4+ memory T cells, CD8+ Tcm, NK_Cells, and Tgd_Cells, and negatively with CMP, Mast_Cells, Macrophage_M2, Melanocytes, and NKT. Additionally, we analyzed immune cell changes between the high-risk and low-risk groups in T1DM (Fig. 9B) and observed a significant decrease in CD4+ memory T cells and CD4+ T cells in the low *CD3D* expression group. These results suggest that *CD3D* expression is closely linked to CD4+ T cell expression in the T1DM and MCI groups.

4. Discussion

T1DM and T2DM are associated with reduced performance in multiple domains of cognitive function [35,36]. Various mechanisms have been shown to contribute to MCI pathology. The immune system has been demonstrated to play an important role in MCI [37,38]. Microarray and sequencing methods have facilitated the exploration of the molecular landscape and potential mechanisms underlying miscellaneous diseases [39]. In light of the co-occurrence of AD with diabetes [40] and the identification of MCI as an early symptom of AD [41], the present

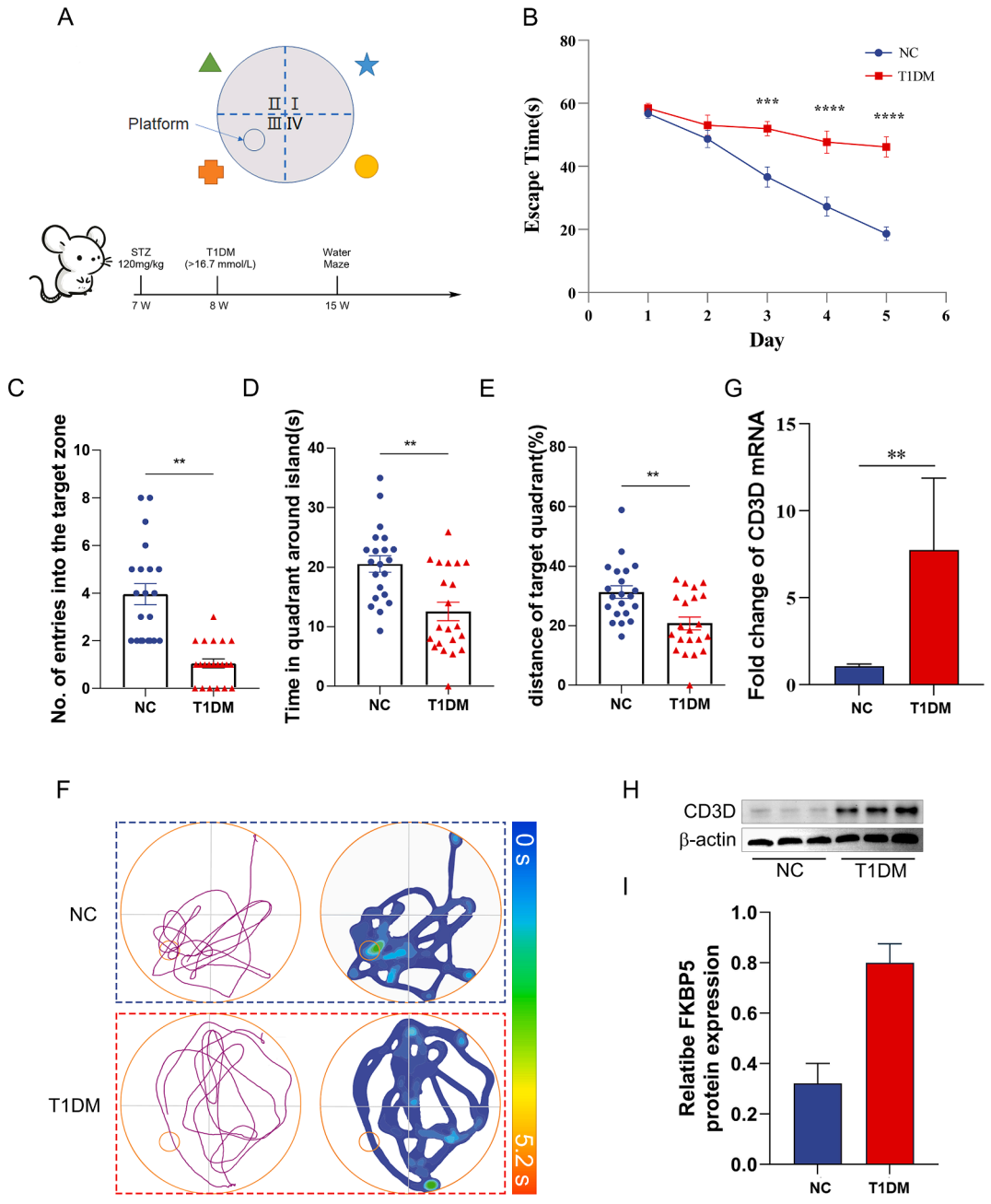


Fig. 7. Morris water maze test and CD3D expression (A) Schematic illustration of the Morris water maze. (B) Learning and memory curve in the first five days. (C) Number of arrivals at the platform (n = 21 per group). (D) Time spent in platform quadrant. (E) Percentage of distance traveled in quadrant of the platform. (F) A route map of mice movements (for density plots, the redder the color, the longer the time). (G) The expression of CD3D in peripheral blood (n = 6 per group).

study entailed a more comprehensive analysis of the immune cell environment and explored immune-associated genes in T1DM and MCI. We used T1DM and MCI datasets and identified a total of 232 upregulated and 378 downregulated common DEGs, respectively, suggesting their potential involvement in the development of T1DM to MCI. The impact of DEGs typically initiates fundamental molecular changes, contributing to a more comprehensive understanding of the underlying mechanisms. The GO and KEGG enrichment revealed that those genes are associated with ribosome function, metabolic signaling pathways, and immune-related pathways.

Machine learning algorithms have been a dominant tool in the data mining field [42,43]. In particular, Feature Selection for Key Gene is useful for identifying biomarkers or genes associated with diseases or phenotypes and helps to evaluate the importance of genes based on their contribution to model prediction. Several studies have described the role of machine learning in disease diagnosis. Yang et al. [44] used machine learning algorithms to develop the T cell-related score (Tstr), a stress response score to predict outcomes in clear cell renal cell carcinoma, and identified *TNFSF13B* as a key Tstr-related gene. Specifically, *TNFSF13B* could serve as a potential biomarker to predict the effects of PD-1 blockade treatment in patients with clear cell renal cell carcinoma. Similarly, machine learning techniques have been used to analyze and verify possible ferroptosis-related genes intricately linked to the progression of membranous nephropathy [45]. Although machine learning algorithms are powerful, they present challenges in gene selection. Overfitting can occur when the model becomes too complex [46], whereas underfitting occurs when the model is too simple, potentially missing significant relationships between genes and outcomes [47] and

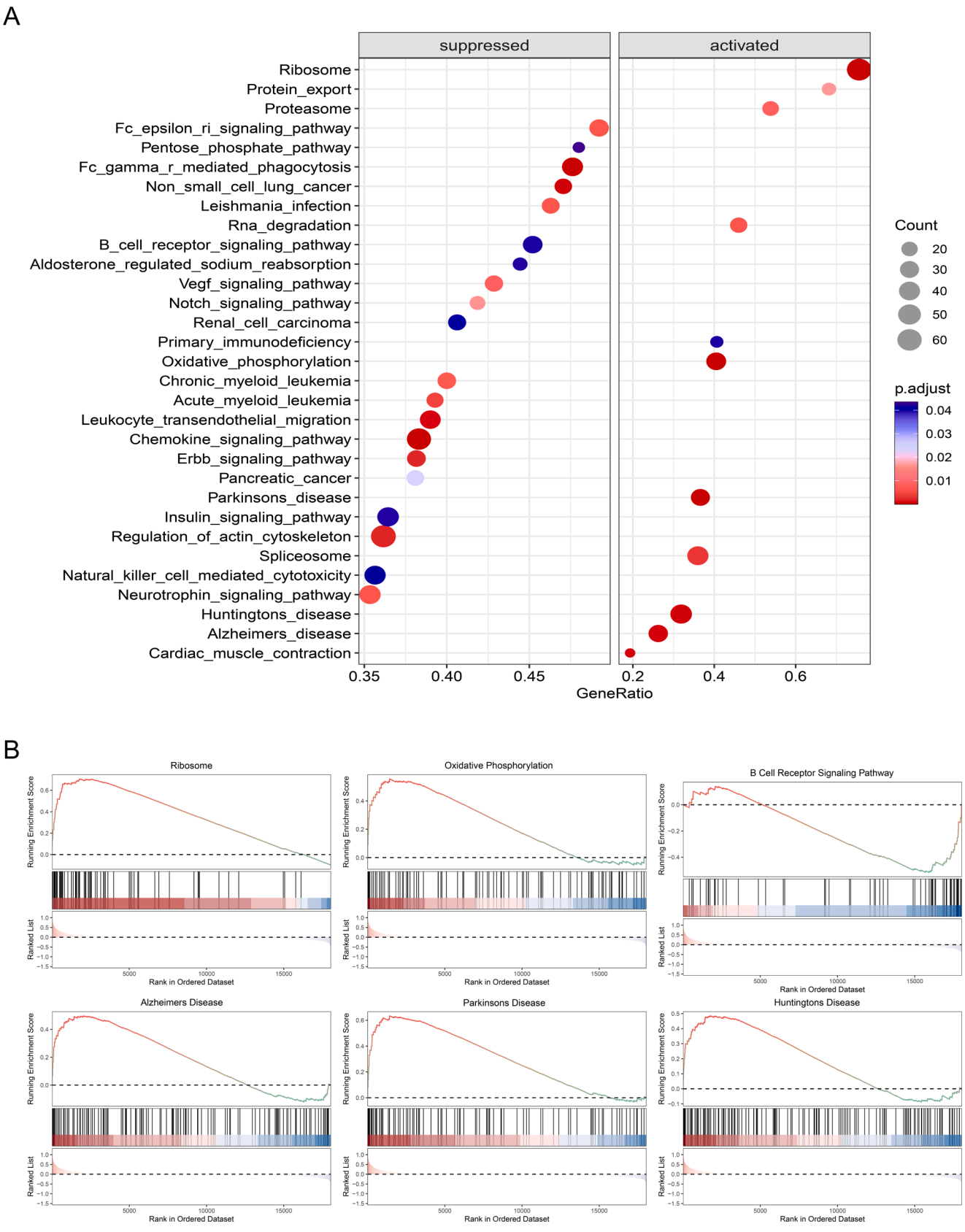


Fig. 8. Function enrichment analysis based on CD3D. (A) GSEA showing the suppressed and activated pathways in the MCI group. (B) GSEA showing the pathways associated with metabolic and neurodegenerative diseases in the T1DM group. GSEA, gene set enrichment analysis.

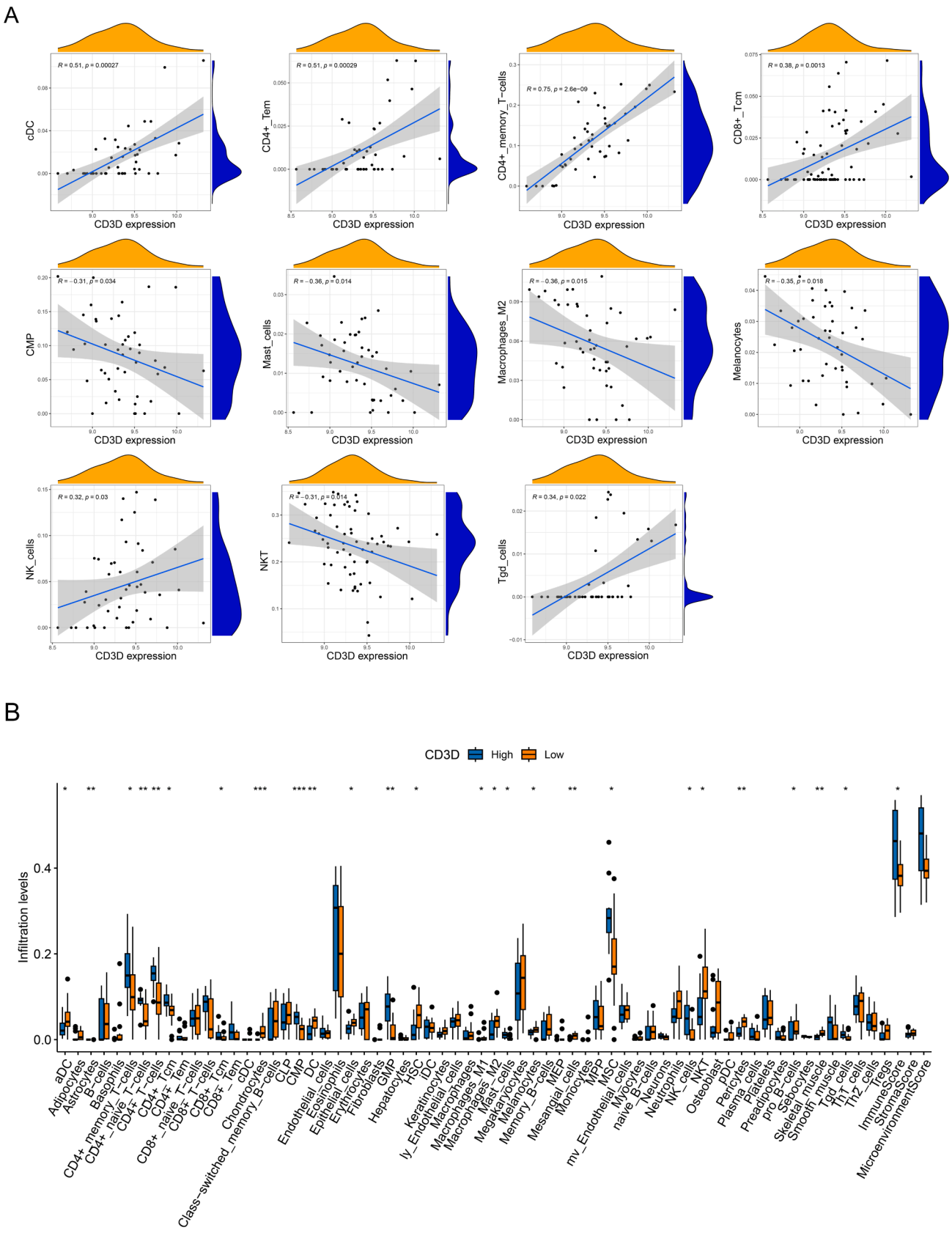


Fig. 9. Changes in immune cell profiles between high-risk and low-risk groups. (A) CD3D expression correlation with various immune cell types, including cDC, CD4+ T cells, CD4+ memory T cells, CD8+ Tcm, CMP, Mast_Cells, Macrophage_M2, Melanocytes, NK_Cells, NKT, and Tgd_Cells. (B) Comparative analysis of immune cell alterations between the high-risk and low-risk groups in T1DM using xCell analysis. *p < 0.05; **p < 0.01.

thereby overlooking key biomarkers. To address these limitations, in the present study, we used a combination of nine distinct algorithms to select nine hub genes, namely, *DAXX, PIK3R2, NR2F6, SLC40A1, PIK3CD, TMSB10, CD3D, CELA1*, and *GMFG*. Next, *CD3D, CELA1,* and *GMFG* as selective for the MCI and T1DM diseases. Using Friends analysis on these nine genes, we found that *CD3D* had the highest score. Unfortunately, GMFG was not detected in the Friends analysis, which indicated that it was not associated with other genes derived from the GO enrichment analysis.

The AUC is a widely used parameter in disease diagnosis models, where the goal is to maximize the classifier's ability to correctly differentiate between diseased and non-diseased individuals. We used the AUC to evaluate the ability to correctly identify diseased individuals and found that the values for those nine marker genes were >0.59, demonstrating that they were accurate and specific in differentiating psoriasis cases from healthy samples. Notably, *CD3D* had the highest AUC values both in T1DM (training set: AUC = 0.730; validation set: AUC = 0.844) and MCI group (training set: AUC = 0.777; validation set: AUC = 0.801). *CD3D* encodes the δ subunit of the transmembrane CD3 antigen complex and forms the T cell receptor/CD3 complex (TCR/CD3 complex) with the other four CD3 subunits for T cell development and signal transduction [48,49]. The immune system is regulated by costimulatory signaling and inhibitory molecules referred to as immune checkpoints [50] that facilitate the initial activation of T cells and elicit responses from effector cells, memory cells, and regulatory T cells (Treg) [51–53]. *CD3D* is an important regulatory molecule of immune checkpoint. *CD3D* plays important roles in nervous system diseases, such as epilepsy [54], depression [55], Primary Sjögren's Syndrome [56], and other nervous system diseases [57–59]. Furthermore, *CD3D* exhibits high sensitivity and specificity in anti-cyclic citrullinated peptide antibody-negative rheumatoid arthritis patients [60]. This suggests *CD3D* as a potential key mediator in the pathogenesis of rheumatoid arthritis and a marker for its diagnosis. This finding underscores the positive impact of *CD3D* in the context of diagnosing immune-related diseases. Nevertheless, data on the function of *CD3D* in the progression of T1DM to MCI remain limited. To address this, we established a murine T1DM model and observed significant deficits in learning and memory abilities. Furthermore, using quantitative PCR and western blotting, we revealed substantial upregulation of *CD3D* in these mice, suggesting a pivotal role of *CD3D* in the transition from T1DM to MCI.

Herein, a single-gene GSEA-KEGG pathway analysis was conducted to explore the potential functions of marker genes. Our results indicated that pathways related to the ribosome, oxidative phosphorylation, and multiple neurodegenerative diseases were significantly associated with marker genes in the diabetes and MCI groups. Furthermore, we analyzed immune cells in patients with T1DM and MCI compared with healthy controls. In the high-risk *CD3D* group, we observed a significant increase in the expression levels of CD4+ T cells, suggesting a close association between *CD3D* and CD4+ T cell function. Similarly, Li et al. [61] reported an increase in CD4+ T cells and macrophages during the progression of T1DM. Brusko et al. [62] evaluated peripheral blood samples from 70 T1DM patients and 37 healthy individuals and revealed that CD4+ T cell expression increased with age. Importantly, Fu et al. [63] found that the percentage of CD4+ Tregs was the highest in the blood of patients with MCI (9.24 %), followed by healthy controls (8.19 %), with the lowest proportion in patients with moderate-to-severe AD-related dementia (7.42 %). In this study, we determined the potential of *CD3D* as a diagnostic marker for MCI resulting from T1DM. These findings indicate that *CD3D* may facilitate the transition from T1DM to MCI by modulating CD4+ T cell function.

Overall, our study revealed that *CD3D* plays a pivotal role in the transition from T1DM to MCI by modulating CD4+ T cell function, positioning it as a potential diagnostic marker for early cognitive decline in T1DM patients. However, this study also has some limitations. First, although preliminary analyses indicated potential correlations between the *CD3D* gene and T1DM, as well as MCI, further comprehensive investigations are warranted, including the substantiation of these findings using additional biological experiments. Second, to enhance the robustness of our study, validation utilizing multicenter clinical data is warranted, which, when combined with our findings, may reveal novel avenues for research.

Ethics approval and consent to participate

All procedures were conducted in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals and approved by the Institutional Animal Care and Use Committees of KunMing Medical University.

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CRediT authorship contribution statement

Dongxu Yue: Visualization, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Runze Wang:** Data curation, Validation. **Yanli Zhao:** Resources, Investigation, Data curation, Methodology, Conceptualization. **Bangxu Wu:** Validation. **Shude Li:** Supervision. **Weilin Zeng:** Supervision. **Shanshan Wan:** Data curation. **Lifang Liu:** Data curation. **Yating Dai:** Data curation. **Yuling Shi:** Data curation. **Ruobing Xu:** Data curation. **Zhihong Yang:** Conceptualization. **Xie Wang:** Writing – review & editing, Conceptualization. **Yingying Zou:** Writing – review & editing, Conceptualization, Funding acquisition, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intimp.2024.113256.

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